# Threat Intelligence Report

## 1. Basic Information

### Malicious IP Address: 51.124.78.146

ASN: AS8075
Organization: Microsoft Corporation
Geolocation: Ireland
Reported Abuse: 23 times from 7 distinct sources
First Reported: December 1, 2020
Most Recent Report: March 18, 2025 (1 month ago)

## 2. Malicious Activity Observed

| Reporter | Date (UTC) | Category | Comment |
|---|---|---|---|
| Admins@Storch | 2025-03-18 | Brute-Force, Exploited Host | Attempted break-in on port 49888 |
| RoboAbuseCheck | 2024-10-24 | Port Scan | Unsolicited connection to port 57909 |
| GOVT of India | 2024-08-08 | IoT Targeted, Exploited Host | Indicators of compromise (IoCs) |
| RoboAbuseCheck | 2024-07-29 | Port Scan | Multiple unsolicited connection attempts to high-numbered ports |

## 3. Indicators of Malicious Behavior

- Unsolicited SYN packets from local host (192.168.100.8) to 51.124.78.146
- No prior communication history, indicating scanning or beaconing
- Part of a pattern — the host contacted multiple suspicious external IPs (possible C2 communication)

## 4. Reputation Services

| Source | Verdict |
|---|---|
| AbuseIPDB | Reported as malicious |
| AlienVault OTX | Flagged with IOCs |
| VirusTotal | Flagged by multiple engines |
| GreyNoise | History of mass scanning |

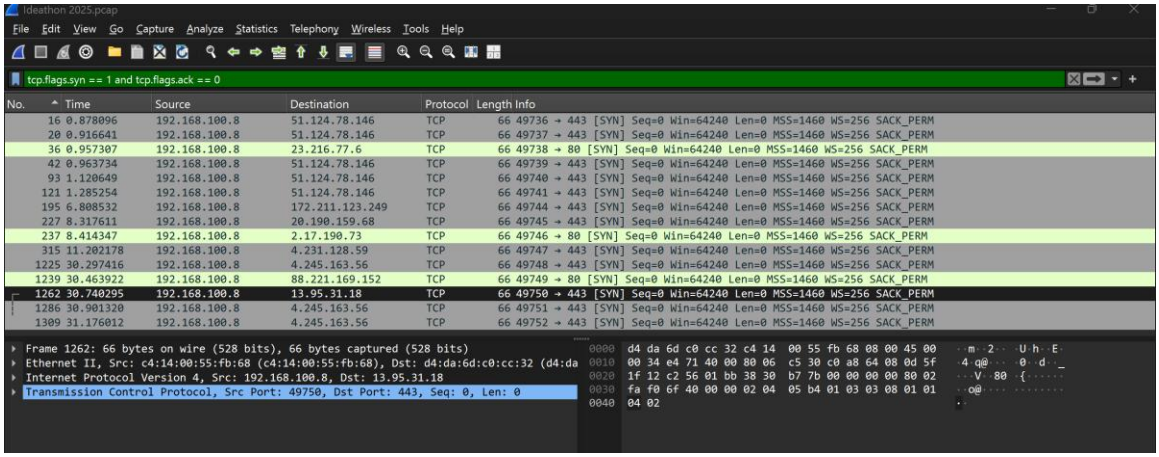
## 5. Risk Assessment

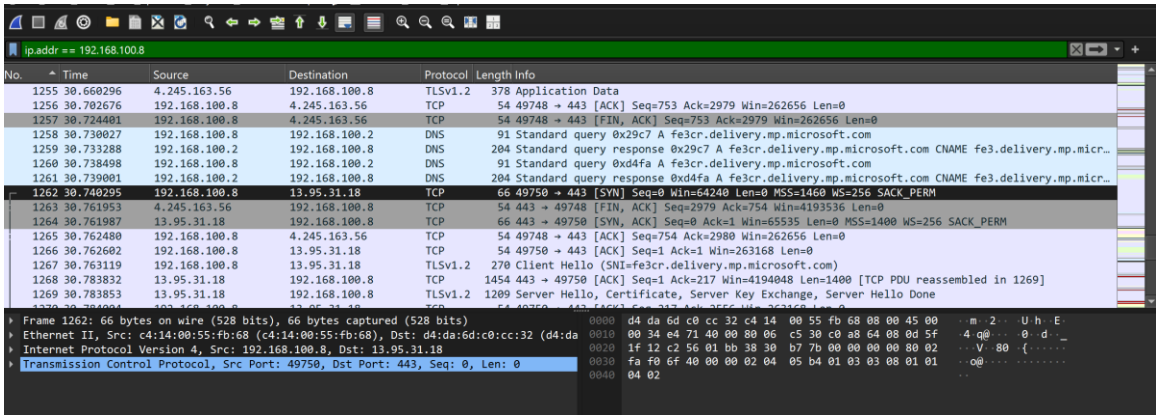
Risk Level: HIGH
Reason: Evidence of:
- Port scanning
- Brute-force attempts
- Potential C2 communications
- Previously exploited history

Recommendation:

Block this IP and investigate systems that communicated with it.

## Analysis of the PCAP file using wireshark:






## Proof of intelligence of the identified malicious IP

*Link: https://www.abuseipdb.com/check/51.124.78.146*

IP Abuse Reports for **51.124.78.146**:

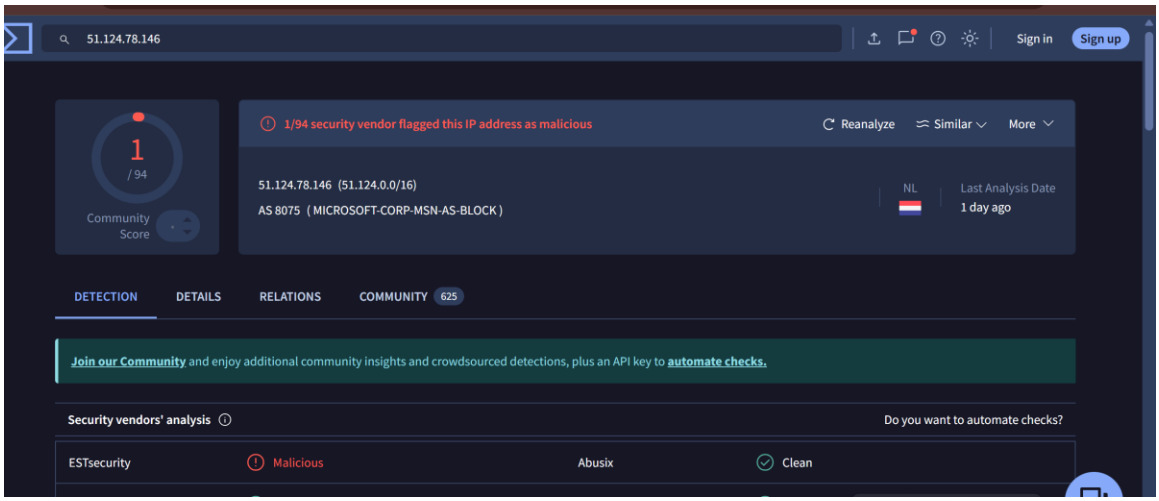
This IP address has been reported a total of **23** times from 7 distinct sources. 51.124.78.146 was first reported on December 1st 2020, and the most recent report was **1 month ago**.

**Old Reports:** The most recent abuse report for this IP address is from **1 month ago**. It is possible that this IP is no longer involved in abusive activities.

| Reporter | IoA Timestamp in UTC | Comment | Categories |
|---|---|---|---|
| ✔ Admins@Storch | 2025-03-18 11:10:13 (1 month ago) | IPS:drop <attempted break-in> dstport=49888 | Brute-Force, Exploited Host |
| ✔ RoboAbuseCheck | 2024-10-24 01:58:38 (6 months ago) | Unsolicited connection attempts to port 57909 | Port Scan |
| TC Saini | 2024-08-08 14:01:37 (9 months ago) | Indicators of compromise (IoCs) reported by the GOVT of India | Blog Spam, Exploited Host, IoT Targeted |
| ✔ RoboAbuseCheck | 2024-07-29 13:58:33 (9 months ago) | Unsolicited connection attempts to ports 33941, 33945, 33947, 38493, 63459, 63460, 63461 | Port Scan |
| ✔ RoboAbuseCheck | 2024-07-21 05:58:52 | Unsolicited connection attempts to ports 53538, 53539, | Port Scan |

| | | | |
|---|---|---|---|
| ✔ RoboAbuseCheck | 2024-07-19 05:58:39 (9 months ago) | Unsolicited connection attempts to ports 49721, 49739, 49772, 49923, 51052, 55513, 56119, 561 ... show more | Port Scan |
| ✔ RoboAbuseCheck | 2024-07-19 03:59:07 (9 months ago) | Unsolicited connection attempts to ports 47825, 47830, 47832 | Port Scan |
| ✔ RoboAbuseCheck | 2024-07-17 10:59:13 (9 months ago) | Unsolicited connection attempts to port 55180 | Port Scan |
| ✔ RoboAbuseCheck | 2024-07-08 13:30:47 (10 months ago) | Unsolicited connection attempts to ports 29047, 29071, 33430, 34799, 36090, 36778, 38230, 384 ... show more | Port Scan |
| ✔ RoboAbuseCheck | 2024-07-08 13:14:42 (10 months ago) | Unsolicited connection attempts to ports 37773, 39571, 41827, 43809, 46716, 46868, 47181, 477 ... show more | Port Scan |
| ✔ ChillScanner | 2022-10-11 13:39:25 (2 years ago) | 1 probe(s) @ TCP(49152) | Port Scan |
| ✔ ChillScanner | 2022-09-22 02:39:13 (2 years ago) | 1 probe(s) @ TCP(49152) | Port Scan |
| ✔ ChillScanner | 2022-09-06 13:39:24 (2 years ago) | 1 probe(s) @ TCP(49152) | Port Scan |

## Virus Total



## TTP Mapping Table

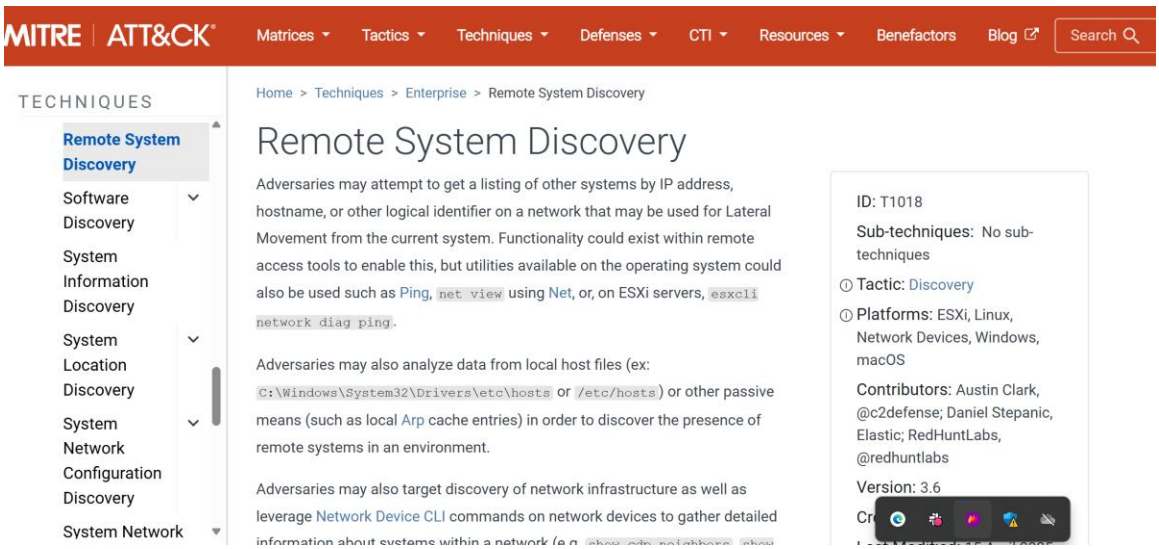
| Task | ID | Name | Use |
|---|---|---|---|
| Initial Access | T1566.001 | Spearphishing Attachment | Malware disguised as a game mod (e.g., 'Free Fortnite cheat.exe') sent via email/Discord. |
| Discovery | T1018 | Remote System Discovery | Attacker scans for vulnerable game servers (e.g., Minecraft, CS:GO). |
| Defense Evasion | T1036.004 | Masquerade File Type | Malware hidden as a game file (e.g., 'steam_api.dll'). |
| Credential Access | T1056.001 | Input Capture: Keylogging | Steals Steam/Epic Games passwords via a keylogger in a fake game launcher. |
| Lateral Movement | T1210 | Exploitation of Remote Services | Exploits a vulnerability in a game server to move to other systems. |
| Command & Control | T1071.001 | Web Protocols | C2 traffic disguised as game update requests (e.g., fake 'Steam Client Update'). |
| Collection | T1113 | Screen Capture | Captures screenshots of in-game purchases (credit card info). |
| Exfiltration | T1048.003 | Exfiltration Over Alternative Protocol | Sends stolen data via game chat (e.g., Discord Webhooks). |
| Impact | T1486 | Data Encrypted for Impact | Ransomware encrypts game saves (e.g., 'Pay $500 to unlock your Skyrim characters'). |

## Proof of concept

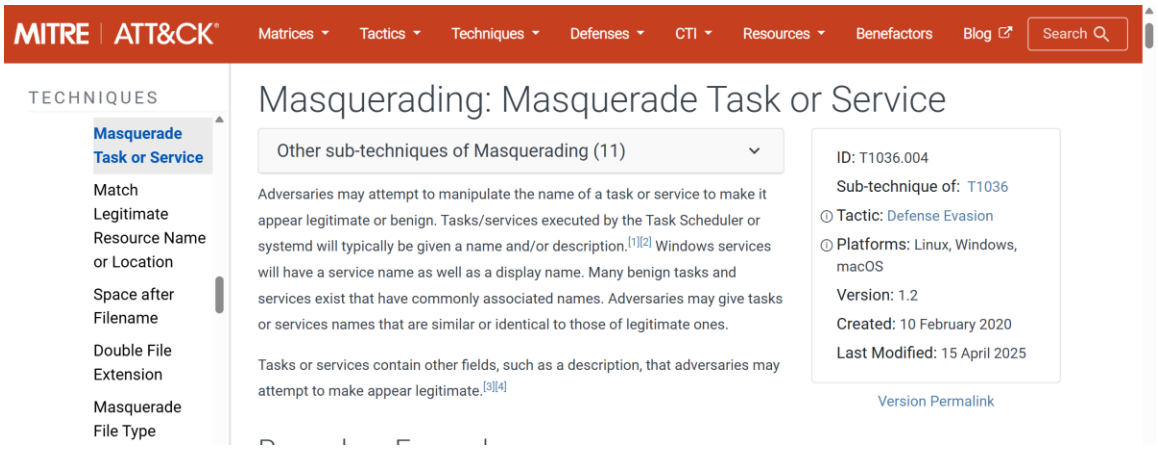
Initial access




Discovery



Defense Evasion



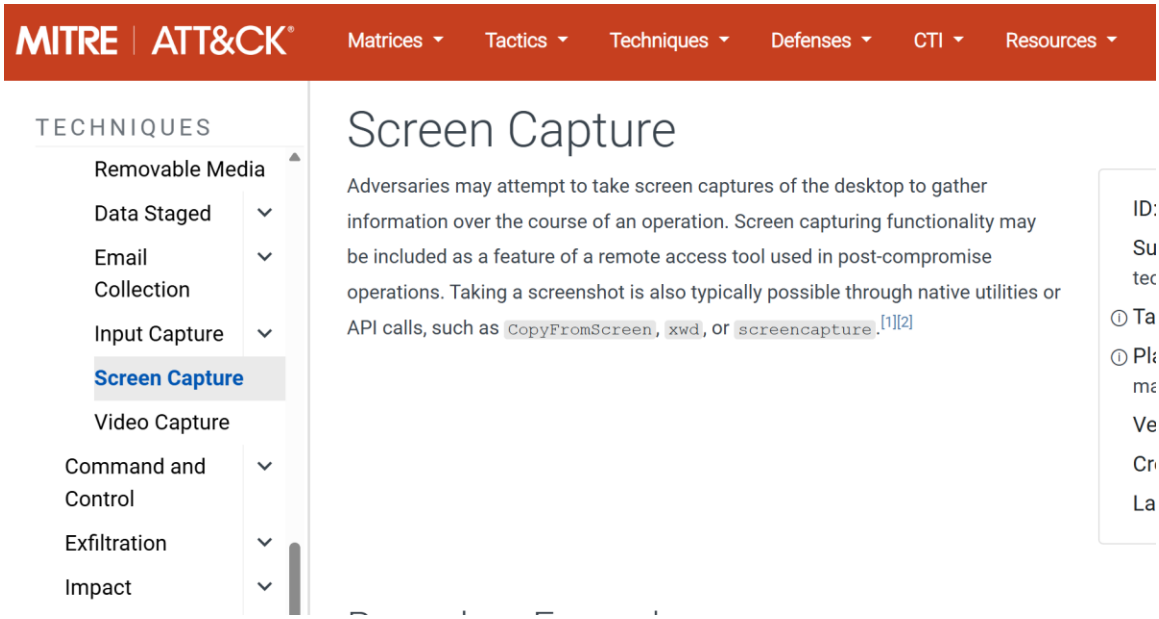

Lateral Movement



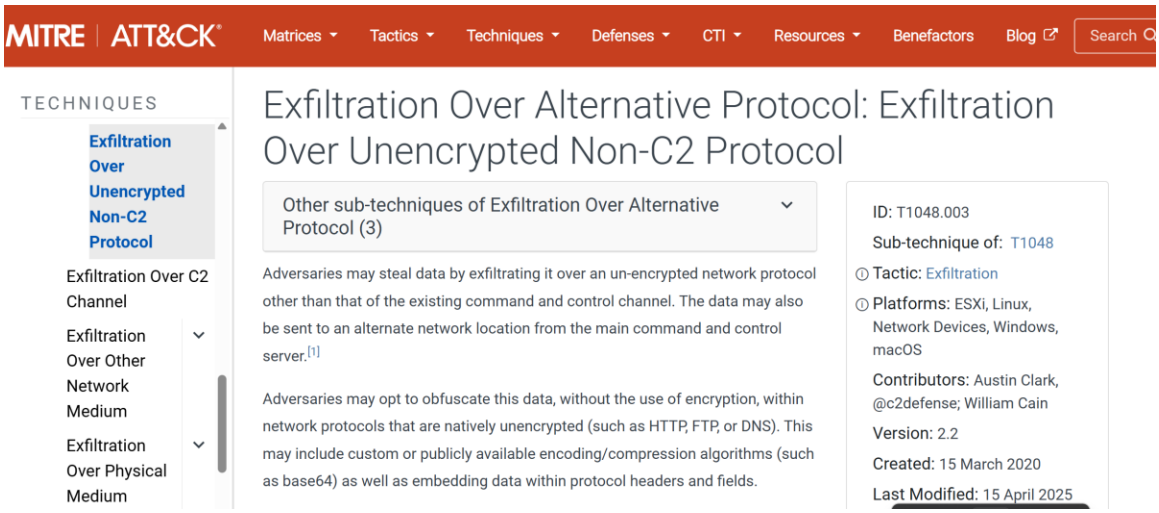
Command and Control



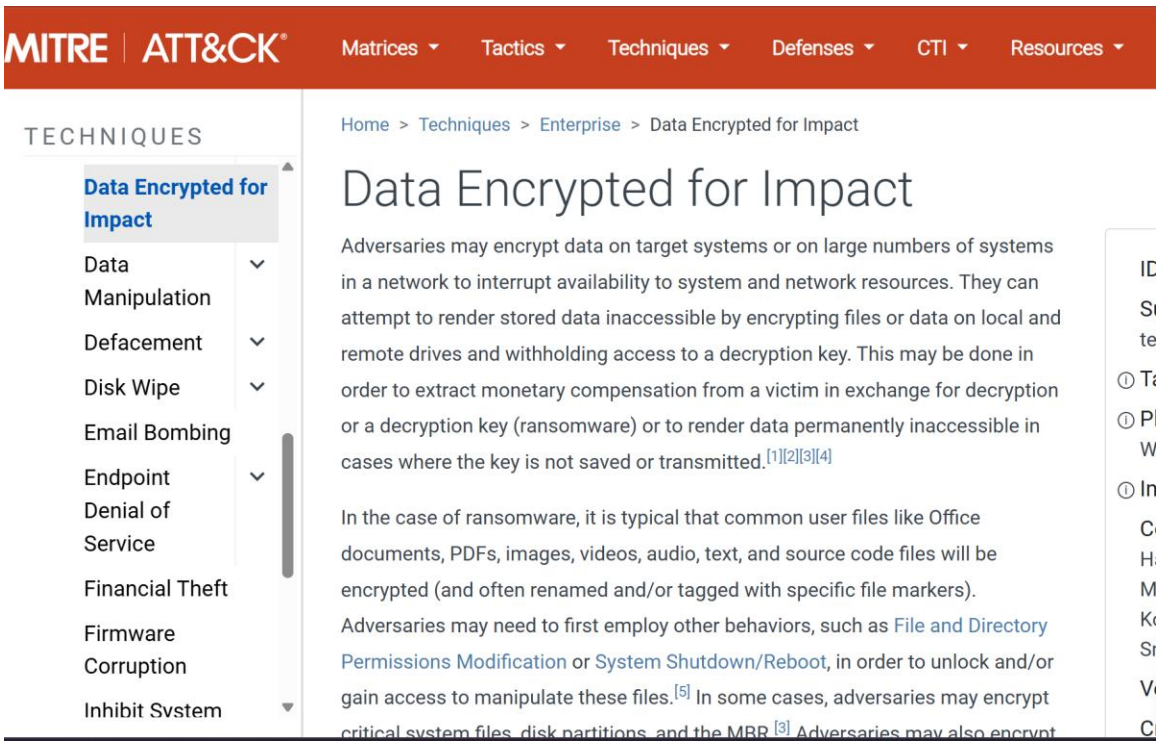

Collection



Exfiltration




Impact



# IOC Analysis Report

## Abstract

This report offers a concise overview of the Emotet malware, a highly adaptive cyber threat. It outlines Emotet's behavior, associated risks, real-world impact, and strategic defense mechanisms. The information is intended to inform cybersecurity strategies in hackathon projects focused on malware detection, incident response, or network hardening.

## Malware Type Identification

The IOC *453257c3494addafb39cb6815862403e827947a1e7737eb8168cd10522465deb* is associated with Emotet, a notorious malware family known for acting as a banking Trojan, spam bot, and a distributor of other malware like ransomware (example; Ryuk).

## Hash Type

This is a SHA-256 hash (64-character hexadecimal string).

## Emotet Overview

First identified in 2014, Emotet began as a banking trojan and has evolved into a modular, polymorphic malware platform capable of delivering additional payloads. It spreads primarily through phishing emails that contain infected attachments (usually Word documents with macros) or malicious links. Once executed, Emotet communicates with remote command-and-control (C2) servers to receive updates and download further malware such as TrickBot or ransomware.

## Key Risks and Characteristics

Emotet is particularly dangerous due to its worm-like behavior and ability to evade traditional antivirus solutions through frequent code modifications. It leverages stolen email threads to increase its phishing success rate. Once inside a network, it moves laterally by brute-forcing credentials or exploiting shared drives (CISA, 2021). The malware has been linked to large-scale data theft, spam propagation, financial fraud, and system disruptions. According to the department of justice (2021), Emotet-related incidents can cost organizations up to $1 million to remediate.

## Sector-Wide Impact

Emotet has severely affected sectors like healthcare, banking, education, and government. In 2021, international law enforcement disrupted the botnet, citing "hundreds of millions" of dollars in global damages and infections affecting critical infrastructure (U.S. Department of Justice, 2021). Hospitals experienced ransomware outbreaks, while financial firms reported data breaches and fraud.

## Protection Strategies

Robust defenses against Emotet require a multi-layered approach. Email filtering and advanced threat protection can block malicious attachments and URLs. Consistent patch

management closes vulnerabilities that Emotet may exploit. Enforcing strong password policies and multi-factor authentication limits lateral movement. Network segmentation ensures that an infected machine cannot spread malware easily to sensitive systems. Offline, encrypted backups are critical for ransomware recovery, and employee training remains one of the most effective long-term defenses (Department of Justice, 2021).

## IOC Utilization in Cyber Defense

Indicators of Compromise (IOCs), including file hashes, malicious IPs, and domains, are essential tools in Emotet detection and response. These can be integrated into SIEM systems to trigger alerts or fed into intrusion detection/prevention systems (IDS/IPS) for real-time blocking. Platforms like VirusTotal, MISP, and national cybersecurity advisories regularly update Emotet IOCs. When incidents occur, IOCs help teams locate infections, isolate affected devices, and contain further spread (FireEye, 2020).

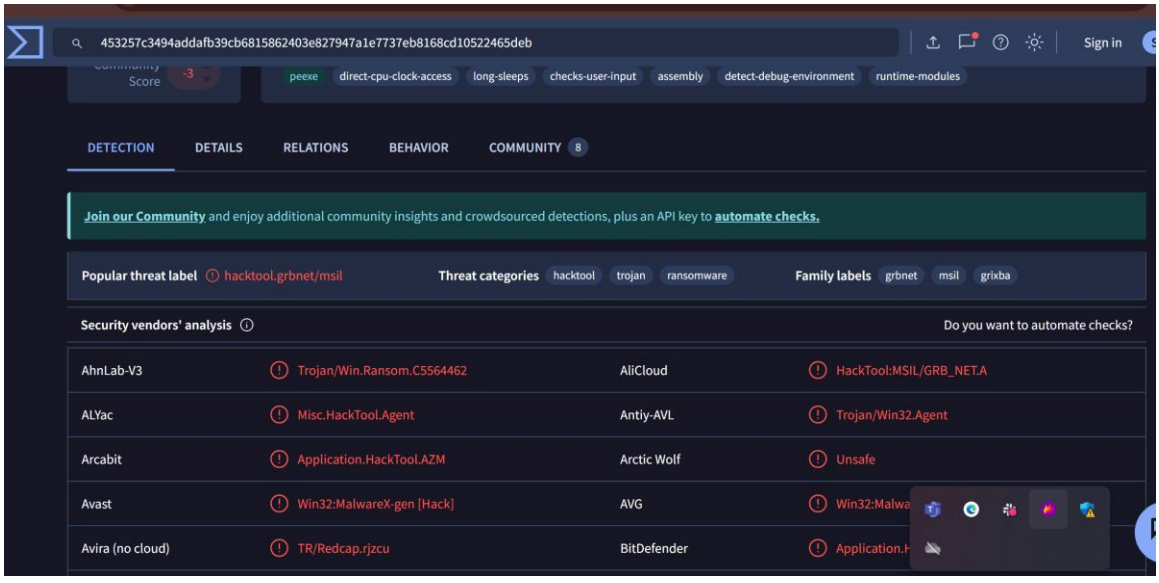
## References


- U.S. Department of Justice. (2021, January 27). Emotet botnet disrupted in international cyber operation [Press release].

https://www.justice.gov/opa/pr/emotet-botnet-disrupted-international-cyber-operation

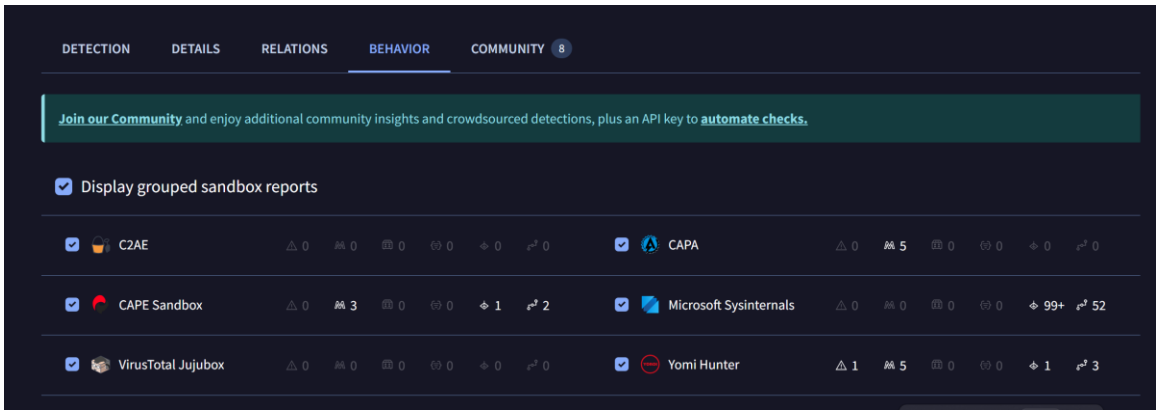

## Proof of concept

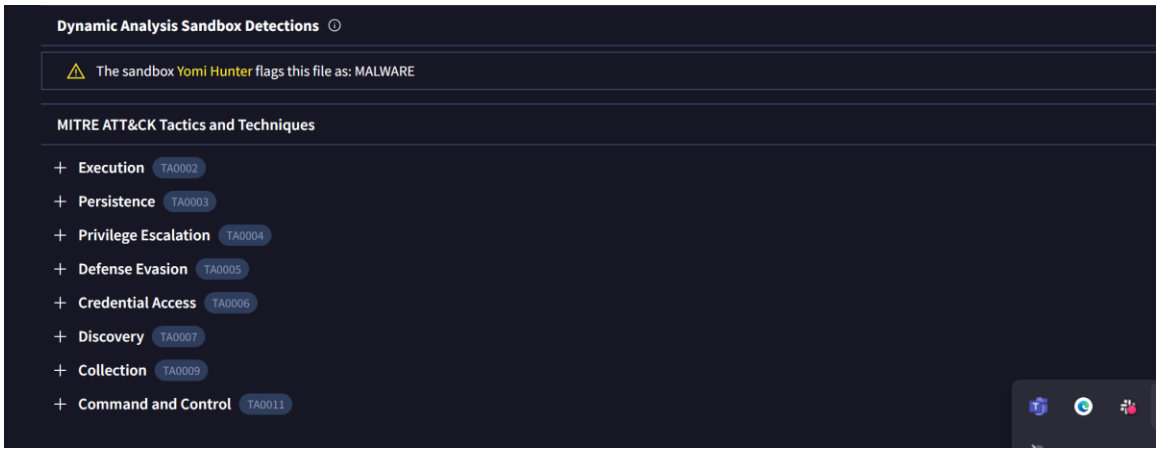
Hash identifier and analysis

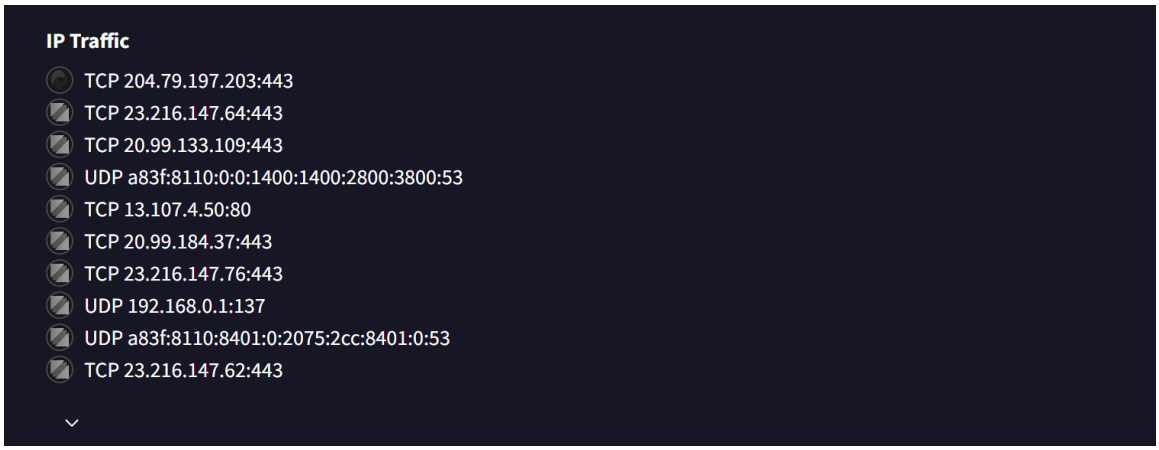


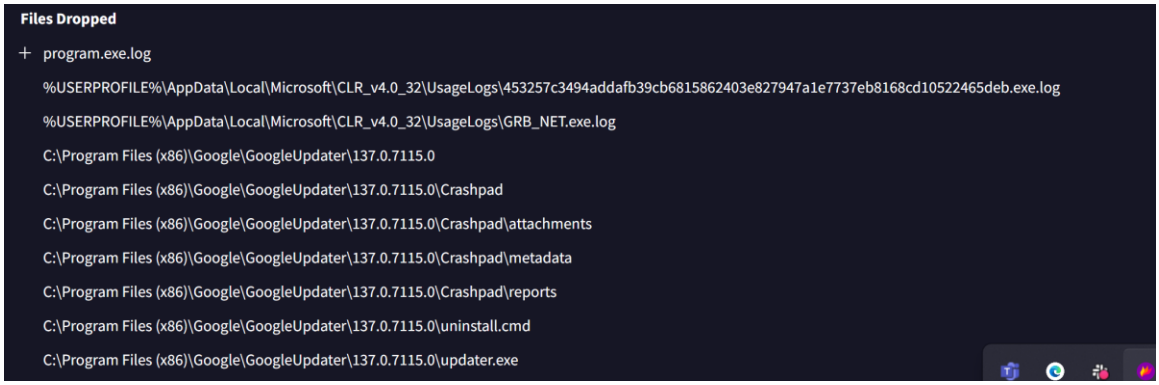
| | | | |
|---|---|---|---|
| Avira (no cloud) | TR/Redcap.rjzcu | BitDefender | Application.HackTool.AZM |
| Bkav Pro | W32.AIDetectMalware.CS | CrowdStrike Falcon | Win/malicious_confidence_100% (W) |
| CTX | Exe.hacktool.grbnet | DeepInstinct | MALICIOUS |
| Elastic | Malicious (high Confidence) | Emsisoft | Application.HackTool.AZM (B) |
| eScan | Application.HackTool.AZM | ESET-NOD32 | A Variant Of MSIL/HackTool.GRB_NET.A |
| Fortinet | W32/Filecoder.PLAY!tr.ransom | GData | Application.HackTool.AZM |
| Google | Detected | Jiangmin | HackTool.MSIL.bcya |
| K7AntiVirus | Hacktool ( 005ab3ac1 ) | K7GW | Hacktool ( 005ab3ac1 ) |
| Kaspersky | HEUR:HackTool.MSIL.GRBNET.gen | Kingsoft | MSIL.HackTool.GRBNET.gen |

## Behavior analysis




Dynamic Analysis Sandbox Detections
The sandbox Yomi Hunter flags this file as: MALWARE
MITRE ATT&CK Tactics and Techniques
Execution TA0002
Persistence TA0003
Privilege Escalation TA0004
Defense Evasion TA0005
Credential Access TA0006
Discovery TA0007
Collection TA0009
Command and Control TA0011


IP Traffic
TCP 204.79.197.203:443
TCP 23.216.147.64:443
TCP 20.99.133.109:443
UDP a83f:8110:0:0:1400:1400:2800:3800:53
TCP 13.107.4.50:80
TCP 20.99.184.37:443
TCP 23.216.147.76:443
UDP 192.168.0.1:137
UDP a83f:8110:8401:0:2075:2cc:8401:0:53
TCP 23.216.147.62:443


Files Dropped
program.exe.log
%USERPROFILE%\AppData\Local\Microsoft\CLR_v4.0_32\UsageLogs\453257c3494addafb39cb6815862403e827947a1e7737eb8168cd10522465deb.exe.log
%USERPROFILE%\AppData\Local\Microsoft\CLR_v4.0_32\UsageLogs\GRB_NET.exe.log
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\attachments
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\metadata
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\Crashpad\reports
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\uninstall.cmd
C:\Program Files (x86)\Google\GoogleUpdater\137.0.7115.0\updater.exe